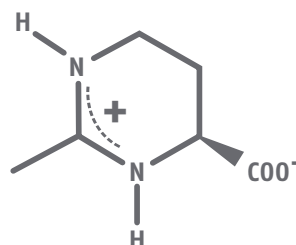


anti-photo aging • 3D cell protection • wrinkle reducing • full spectrum UV and visible light protection • calming • membrane stabilizing • age prevention • DNA protection • long term moisture lock-in • skin barrier repairing • anti-irritation

Ectoin®

ultimate protection active



100% natural and powerful cosmetic active with a very broad spectrum of benefits and applications.

Imagine extreme habitats with incredible dryness, unbelievable heat, very low temperatures or lethal salinity... Where no life can exist – a small stress protection molecule makes the difference: Ectoin®.

The amino acid derivate was discovered as a self-defense and survival substance of microorganisms. They use Ectoin® to protect themselves against the harshness of their environments like salt lakes, hot springs, ice layers or deserts.

bitop
Extremolytes for Life



Boost of skin's self defense

Test protocol:

- » in vitro study
- » human keratinocytes pre-treated with 1% Ectoin®
- » expose to heat-shock for 60 minutes
- » produced stress proteins (HSP 72/72) marked with fluorescent antibodies

Results

**Ectoin® boosts the supply of protection proteins (HSPs).
Skin cells can protect themselves up to 3 times faster!**

Human skin cells produce heat-shock proteins (HSP) under external stress like UV-radiation or other physical and chemical stress factors. Heat-shock proteins recognize, stabilize and metabolize damaged proteins in the skin. They are an important part of the skin's self defense system.

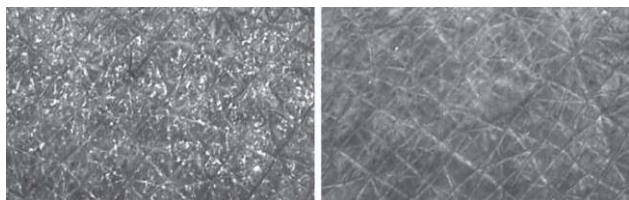
Wrinkle reducing properties – in vivo study

Clinical Trial: test protocol

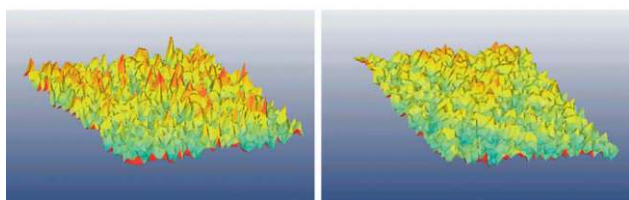
- » double-blind, intra-individual, placebo-controlled study
- » 30 female volunteers (30-60 years old) with normal, dry and sensitive skin using cream containing 2% Ectoin® 2x/day for 28 days and placebo cream
- » skin surface measurement by SELS method, Visioscan at day 0 and day 28

Results

Visualization of the smoothing effect by Visioscan



3D representation of wrinkle depth by SELS method



After 4 weeks of treatment with Ectoin®, a visible and significant reduction of the skin micro relief is observed. Deep structures of the skin were smoothed. The skin looks healthier and younger. 100% of the subjects demonstrated a significant effect.

23% reduction of wrinkle volume after 4 weeks!

Skin elasticity improvement – in vivo study

Clinical Trial: test protocol

- » double-blind, intra-individual, vehicle-controlled application test
- » 30 female volunteers (30–60 years old) with normal, dry and sensitive skin using cream containing 2% Ectoin® and placebo 2x/day for 28 days
- » visco-elasticity & biologic elasticity measurement with Cutometer® (in AU) at day 0 and day 28

Results

Positive influences on skin elasticity are accompanied by a decrease (▼) of parameter visco-elasticity.

Visco-elasticity ▼	day 0	day 28	%	results
2% Ectoin®	0,12	0,11	- 7,5	improvement
vehicle	0,11	0,12	+ 3,5	deterioration
untreated control	0,14	0,15	+ 9,1	deterioration

Improvement of skin elasticity by 7,5%!

Whereas the untreated control and the vehicle treated skin changed for the worse with regard to skin elasticity after 4 weeks treatment, only Ectoin® treated skin showed significant improvement of visco-elasticity and biologic elasticity.

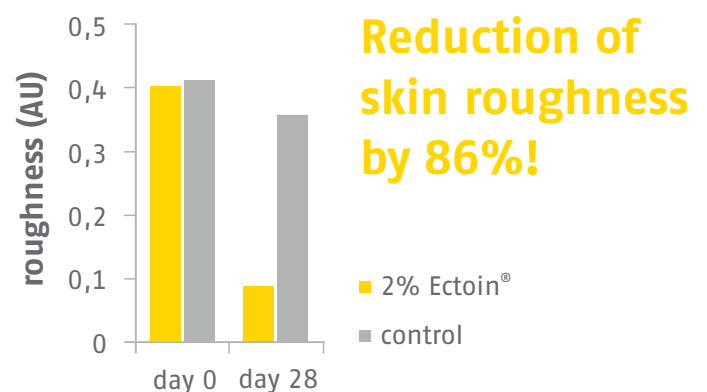
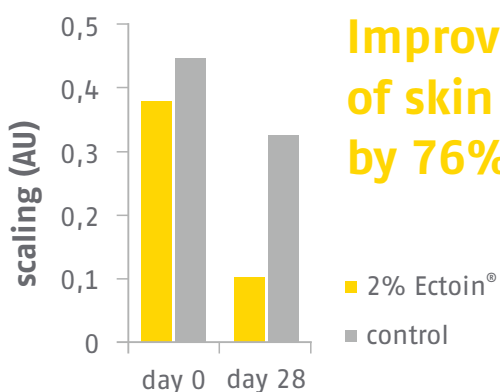
Improvement of scaling and roughness – in vivo study

Clinical Trial: test protocol

- » double-blind, intra-individual, vehicle-controlled application test
- » 30 women (30–60 years), normal, dry & sensitive skin
- » cream with 2% Ectoin®, placebo 2x/day for 28 days
- » skin roughness and scaling measured with Visioscan®

Results

Skin scaling and roughness was significantly decreased after 28 days.



Protection against photo-aging

Ectoin® protects the skin from damage and accelerated skin aging caused by UVA- and UVB-radiation as well as visible light in a number of different ways and at different cell levels.

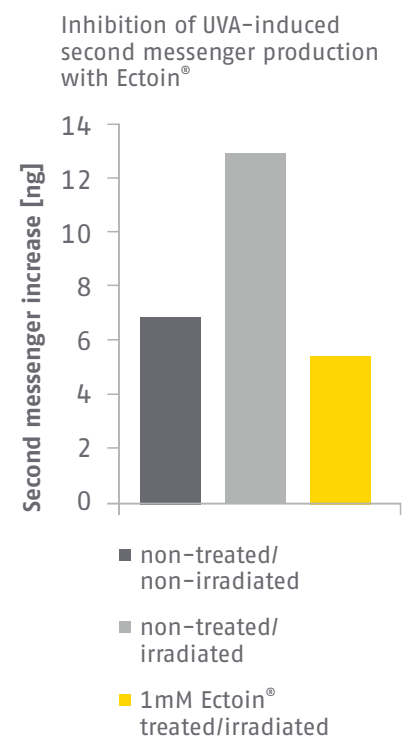
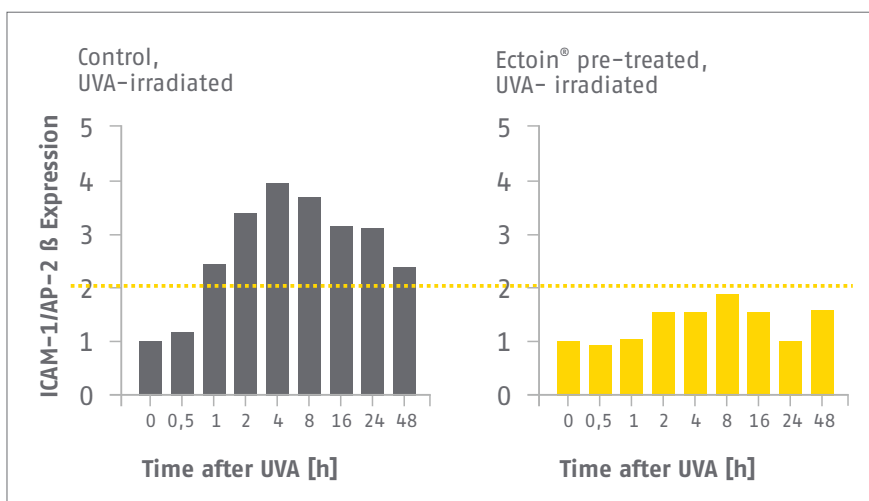
Inhibition of UVA-induced second messenger release, activation of AP-2 and ICAM-1 expression

Test protocol:

- » in vitro with human keratinocytes, 24 hours pre-incubation of keratinocytes with 1 mM Ectoin®
- » irradiation with 30 J/cm² UVA for 1 hour
- » determination of second messenger release, AP-2 activation and ICAM-1 expression

Results

Ectoin® completely inhibits UVA-induced second messenger release, the activation of transcription factor AP-2 and the expression of ICAM-1!



Measured parameters of UVA-stress model have shown:
Ectoin® is capable to totally inhibit the cell damaging process induced by UVA-irradiation.

UVA and visible light protection

Photo protective capacity of Ectoin® against the combination of visible light & UVA-radiation and visible light was measured by comet assay.

Test protocol:

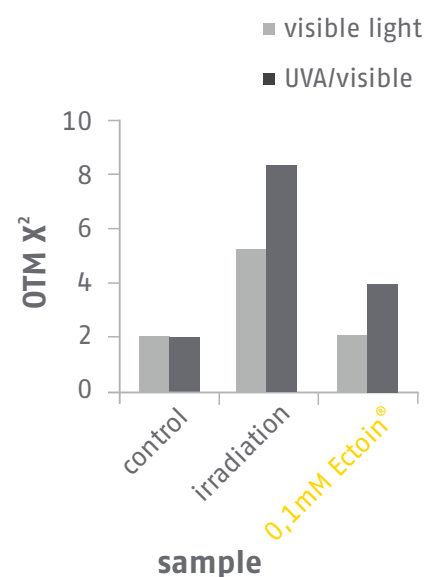
- » pre-treatment of keratinocytes with Ectoin® (0,025, 0,05 and 0,1mM)
- » keratinocytes were irradiated by combined light (UVA/visible) and visible light (dose of 15 J/cm²)

Result

All tested concentrations showed a highly significant level of photo protection against the combination UVA/visible and visible light.

Ectoin® does not adsorb UV radiation, but protects skin cells directly. It brings anti-photo aging on to the cellular level. Ectoin® completes sunscreen and day care formulations for a full spectrum protection against UVA-radiation and visible light damage.

**Level of photo protection:
92,7% (against visible light)
and 68,9% (UVA/visible)!**



Block of UVA-induced DNA damage

UVA-radiation is known to cause mutations and damage of mitochondrial DNA, which leads to premature photo aging.

Test protocol:

- » fibroblasts were pretreated with 1mM Ectoin® over 24h
- » human dermal fibroblasts were exposed 3 times per day to 8 J/cm² on 4 consecutive days over 3 weeks.
- » the mutations of the mitochondrial DNA were measured with polymerase chain reaction.

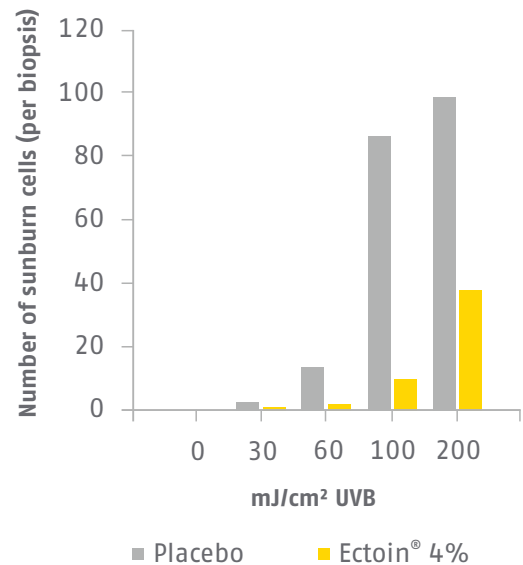
Results

Pre-treatment with Ectoin® was sufficient to completely inhibit UVA-radiation induced mitochondrial DNA mutation!

Reduction of sunburn cells

Test protocol:

- » in vitro study
- » 24 hours pre-treatment with 4% Ectoin® or placebo
- » UV-radiation of organotypic skin (Skinethic®)
- » histological proof of sunburn cells



Result

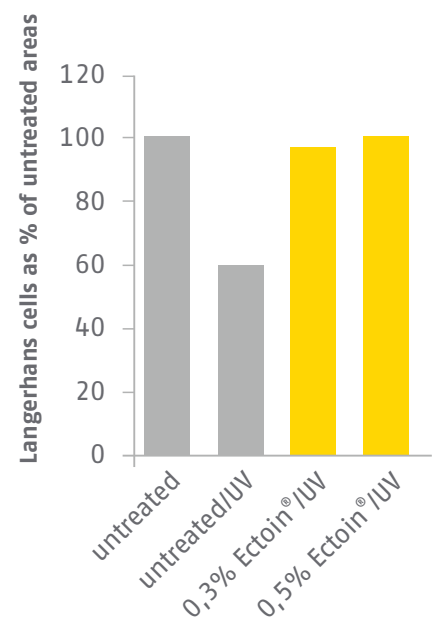
Ectoin® significantly reduces the number of sunburn cells. It protects the skin cells against harmful effects and damages induced by sun exposure.

Dramatic reduction of sunburn cells!

Protection of Langerhans cells – in vivo study

Clinical trial: test protocol

- » in vivo study
- » treatment twice daily with 0, 0.3% or 0.5% Ectoin® for 14 days
- » UV-stress (1.5 MED) of human forearm
- » skin preparation and histological proof of Langerhans cells



Results

Ectoin® protects the number of Langerhans cells in UV-irradiated human skin and thus reduces UV-induced cellular immune-suppression.

Ectoin® protects the skin from premature photo aging by guarding Langerhans cells from UV-induced damage and decreasing!

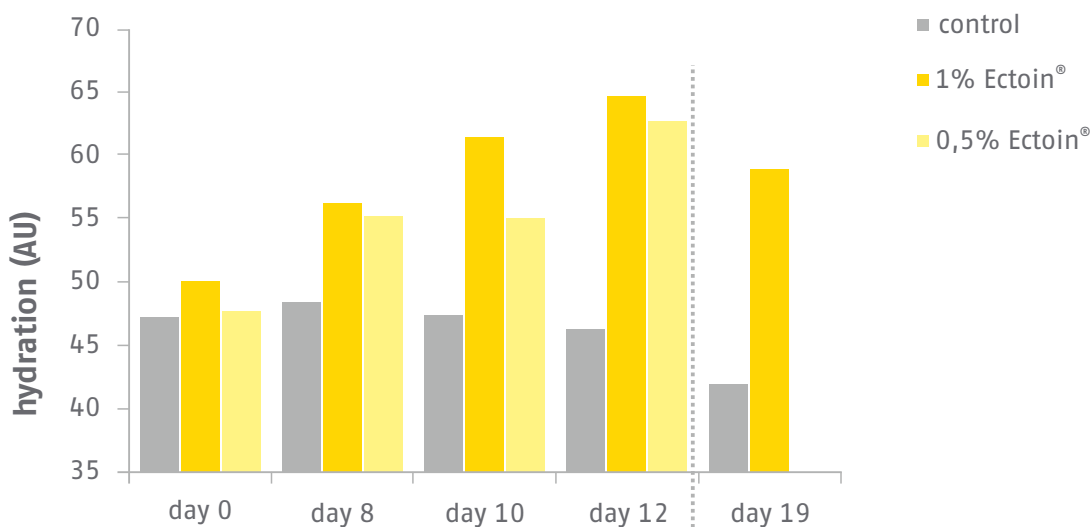
Moisture lock-in (7days) – in vivo study

Clinical trial: test protocol

- » double blind, placebo-controlled application test
- » 5 female volunteers with normal to dry skin
- » formulation with 0%, 0.5% and 1% Ectoin®, applied twice daily for 12 days on forearm
- » skin hydration was measured from day 8 to day 12
- » on day 12, application of Ectoin® was stopped for 7 days
- » at day 19: detecting the skin hydration again and for final result

Results

During the test, the hydration in the skin increased up to 200% compared to the placebo treated skin. Although the topical application was stopped on day 12 the hydration status was preserved for further 7 days.



Increase of skin hydration up to 200% and preservation of hydration status for 7 days!

Ectoin® restores and regulates moisture in the skin by binding water and forming nourishing hydration shells around skin cells.

It has been shown that already after one single application, Ectoin® containing formulations maintain a significantly higher degree of skin moisture than untreated or placebo-treated skin.

Product description

Applications

- » skin care (face, body)
- » baby care
- » sun care
- » hair care

explicitly suggested for ultra sensitive and allergic skin

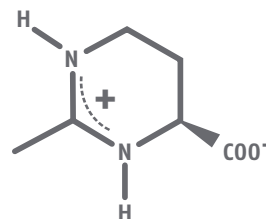
Formulation guidelines

- » level of use 0,3–2%
- » no incompatibility with other cosmetic active substances
- » can be added directly to aqueous phase prior to emulsification

100% safe
GMO-free
China compliant
non allergenic
Ecocert (applying for)
preservative free

Appearance	white crystalline powder, odorless
INCI	Ectoin
Purity	≥ 96% (≤ 4% Hydroxyectoin)
Solubility	excellent solubility in water (390g/l), Glycerol, Propylene glycol
Chemical name	(4S)-2-methyl-1,4,5,6-tetra- hydroypyrimidine-4-carboxylic acid
Molecular formula	C ₆ H ₁₀ N ₂ O ₂
CAS	96 702-03-3
Molar mass	142.16
Production process	biotechnological fermentation / natural origin
Shelf life	5 years
Stability	pH 1–9, temperature up to 80°C, light and oxygen stable
Storage conditions	dry place, temperature from 15° – 25°
Pack sizes	1kg, 5kg, 25kg

Ectoin®
ultimate protection active



bitop

Extremolytes for Life

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